Beyond the Bias Loop: Dynamic Content Distribution Model for Satisfaction-Driven Recommendation*

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Abstract

Personalized recommendation systems optimizing for short-term CTR lock users into narrow content zones—the *bias loop*. We propose a fatigue-aware model on a four-stage spectrum (Z_1–Z_4) with engagement-weighted F, user-type-adaptive λ, and Mere Exposure Effect grounding; scenario analysis projects dead-cross at T2 ($E[S] \approx 6.90$) and 1.46× recovery at T3.

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